

Statistics

Topic 8

- Definition: Methods of collection, organization and analysis of numerical information are collectively called **statistics**. Pieces of numerical and non-numerical information are called **data**. In order to collect data, you need to observe or measure some property; this property is called a **variable**.
- Definition: A variable is **qualitative** if it is not possible for it to take a numerical value. A variable is **quantitative** if it can take a numerical value. A quantitative variable which can take any value in a given range is called a **continuous** variable. A quantitative variable which has clear steps between its possible values is called a **discrete** variable.
- Types of Statistics: **Descriptive Statistics** comprises those methods concerned with collection and describing a set of data so as to yield meaningful information.
 - Example: Summarized large amounts of data collected from the pool games of 2011 cricket world cup to provide immediate meaningful information concerning the performance of each team.

Group A									
Teams	Mat	Won	Lost	Tied	N/R	Pts	Net RR	For	Against
Pakistan	6	5	1	0	0	10	+0.758	1312/275.1	1155/288.0
Sri Lanka	6	4	1	0	1	9	+2.582	1336/218.4	882/250.0
Australia	6	4	1	0	1	9	+1.123	1181/218.5	1030/241.0
New Zealand	6	4	2	0	0	8	+1.135	1257/241.3	1156/284.0
Zimbabwe	6	2	4	0	0	4	+0.030	1288/288.0	1189/267.4
Canada	6	1	5	0	0	2	-1.987	1054/295.3	1582/284.5
Kenya	6	0	6	0	0	0	-3.042	932/300.0	1366/222.1
Group B									
Teams	Mat	Won	Lost	Tied	N/R	Pts	Net RR	For	Against
South Africa	6	5	1	0	0	10	+2.026	1595/292.3	1028/300.0
India	6	4	1	1	0	9	+0.900	1673/282.3	1505/299.4
England	6	3	2	1	0	7	+0.072	1600/298.4	1576/298.1
West Indies	6	3	3	0	0	6	+1.066	1299/262.2	1138/292.5
Bangladesh	6	3	3	0	0	6	-1.361	1017/290.2	1276/262.2
Ireland	6	2	4	0	0	4	-0.696	1393/296.5	1595/296.0
Netherlands	6	0	6	0	0	0	-2.045	1182/300.0	1641/274.1

- **Inferential Statistics** comprises those methods concerned with analysis of a subset of data leading to predictions or inferences about the entire set of data.
 - Example: Suppose we collected data for 30 years regarding the average rainfall in the month of July in Lahore, and the amount came to be 3.3 centimeters. We make the inference that next year in the month of July we can expect between 3.2 and 3.4 centimeters of rain.

Where can Statistics Contribute?

□ Business and Industry ○

Manufacturing

- Improve product quality
- Increase efficiency of processes ○ Marketing
- Conduct sample surveys
- Determine product viability
- Estimate advertisement effectiveness ○ Engineering
- Make consistent product
- Predict product life

○ Banking

- Estimate the risk of a company defaulting on loan
- Determine effective asset allocation for portfolios

• Health and Medicine ○ Epidemiology

- Calculate cancer incidence rates
- Monitor disease outbreaks
- Study risk factors for various diseases ○ Public Health
- Design community efforts
- Education programs ○ Pharmacology
- Drug discovery, development, and approval
- Ensure validity of results in clinical trials

○ Genetics

- Identify potential indicators for specific diseases or traits
- Test gene modification for treatment of diseases

• Natural Resources ○ Agriculture

- Evaluate differences in crop management
- Evaluate the best combinations of fertilizers, pesticides and densities of planting.

○ Ecology

- Study changes in local and global climate
- Develop strategies to improve the environment
- Study the impact of new industrial plants on surrounding ecology ○

Geography

- Evaluate the amount of rainfall one can expect for a given area based on longitude, latitude and distance from the sea.

• Social and Natural Sciences ○ Physics

- Determine when an increase in the density of cosmic rays signals the presence of a supernova

- Conducting tests to determine existence of new particles.
- Chemistry
 - Predict shape of large molecules
 - Analysis of mass spectrometry data ○ Biology
 - Seek to better understand why insects cluster
 - Identify genes related to a particular disease ○ Sociology
 - Estimate the chances of a major war in the next five years
 - Study the increase in rates of marriage failure
 - Determine the characteristic of prisoners to study risk of them repeating criminal behavior. ○ Psychology
 - Study the effects of narcotics on schizophrenia
 - Determine if the existence of extrasensory perception can be demonstrated
 - Evaluate the relationship between shyness and loneliness ○
- Anthropology
 - Determine the age of an archaeological site
 - Analyze the percentage difference in body fat between urban and rural dwellers in Pakistan ○
- Zoology
 - Evaluate the differences in behavior of caged animals when they are outdoors and when they are indoors
 - Determine what techniques are more effective in counting a given species of bird.
- Education
 - Develop effective teaching strategies
 - Research appropriate and informative evaluation (testing) instruments ○ Identify risk factors for bullying, dropping out, failing
 - Identify factors contributing to a decrease/increase in student achievement
 - Study the proportion of graduates of various programs subsequently employed in their field of study
- Government Media and Law
 - Government Agencies
 - Design and implement effective sampling strategies
 - Estimate the unemployment rate
 - Track and report changes in the economy
 - Regulations on stock trading, drug approval, pollution ○ Journalism
 - Effective communication of statistical ideas to a broad audience
 - Participation in education efforts ○ Expert Witness
 - Testify in court cases involving DNA evidence, salary discrepancies, discrimination, and disease clusters

- Definitions: A **population** is defined as the set of all possible members of a stated group. A cross-section of the returns of all of the stocks traded on the New York Stock Exchange (NYSE) is an example of a population.
- Definition: A **sample** is defined as a subset of the population of interest. Once a population has been defined, a sample can be drawn from the population, and the sample's characteristics can be used to describe the population as a whole.
- Definition: A measure used to describe a characteristic of a population is referred to as a **parameter**.
- Definition: It is frequently too costly or time consuming to obtain measurements for every member of a population, if it is even possible. In this case, a sample may be used. In the same manner that a parameter may be used to describe a characteristic of a population, a **sample statistic** is used to measure a characteristic of a sample.
- Measurements Scales:
 - **Nominal scale:** Observations are classified or counted with *no particular order*. It consists of assigning items to groups or categories. *No quantitative information* is conveyed and *no ordering* of the items is implied. Nominal scales are therefore *qualitative* rather than quantitative.
 - Religious preference, race, and gender are all examples of nominal scales.
 - **Ordinal scale:** All observations are placed into separate categories and the *categories are placed in order with respect to some characteristic*. Differences between values makes no sense.
 - Political parties on left to right spectrum given labels 0, 1, 2; restaurant ratings, etc, are examples of ordinal scales.
 - **Interval scale:** This scale provides *ranking* and assurance that differences between scale values are equal. *Difference makes sense*, but ratio doesn't; and there is *no natural zero*.
 - temperature (C,F) and dates are examples of interval scale
 - **Ratio scale:** These represent the strongest level of measurement. In addition to providing ranking and equal differences between scale values, ratio scales have a *true zero point as the origin*.
 - Height, weight, age and length are all examples of ratio scale.

Representation of Data

Topic 9

- Definition: A **frequency distribution** is a tabular presentation of statistical data that aids the analysis of large data sets. Frequency distributions summarize statistical data by assigning it to specified groups, or intervals.

Three steps are required to construct a frequency distribution:

- Define the intervals.
- Tally the observations.
- Count the observations.

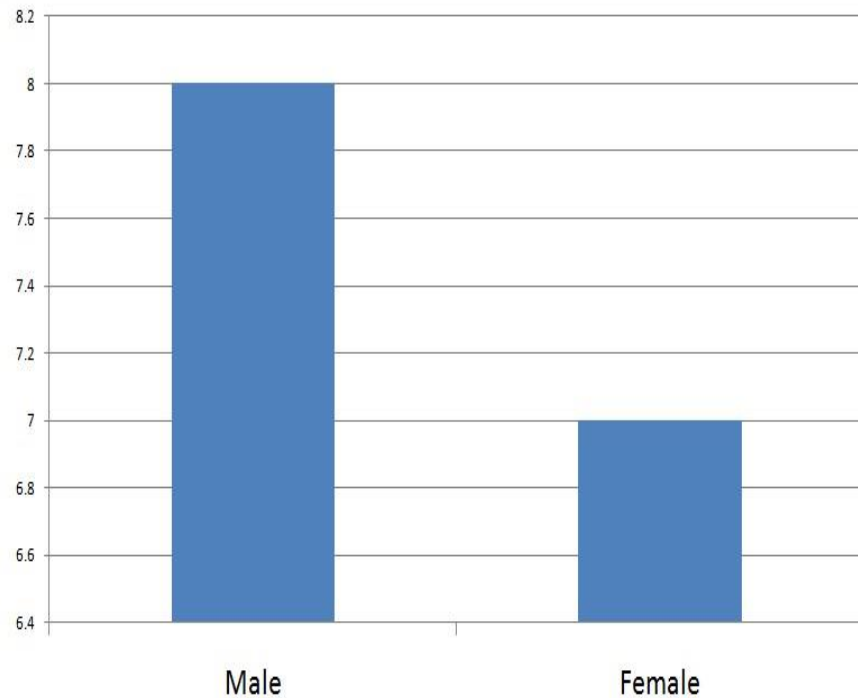
- Definition: **Relative frequency** is calculated by dividing the frequency of each interval by the total number of observations. Simply, relative frequency is the percentage of total observations falling within each interval.

- Definition: **Cumulative Frequency** is calculated by summing the frequencies starting at the lowest interval and progressing through the highest. Cumulative frequency for any given interval is the sum of the frequencies up to and including the given interval.

□ Example:

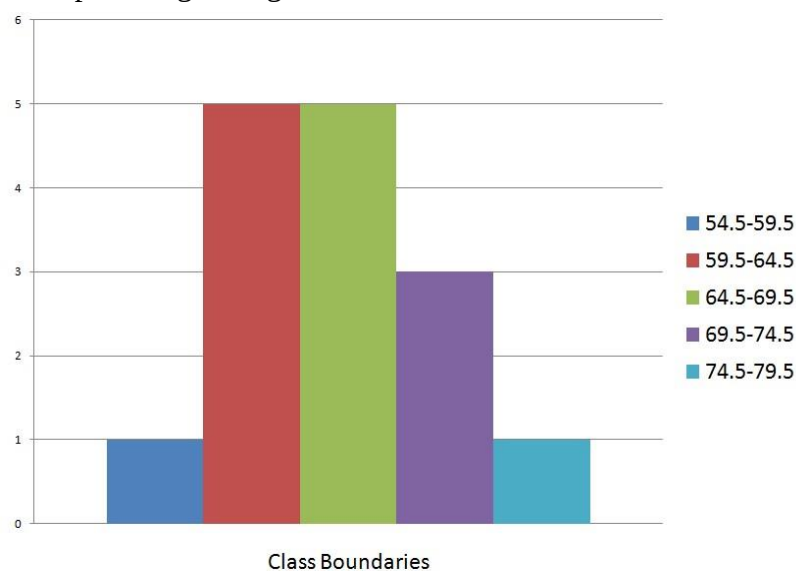
Height Range	Frequency	Cumulative Frequency	Relative Frequency	Cumulative Relative Frequency
4.5 – 5.0 ft	25	25	0.25	0.25
5.0 – 5.5 ft	35	60	0.35	0.60
5.5 – 6.0 ft	29	89	0.29	0.89
6.0 – 6.5 ft	11	100	0.11	1.00

- Definition: A **Bar chart** graphically represents the data sets by representing the frequencies as heights of bars.
 - Example: Genderwise groupings of



students.

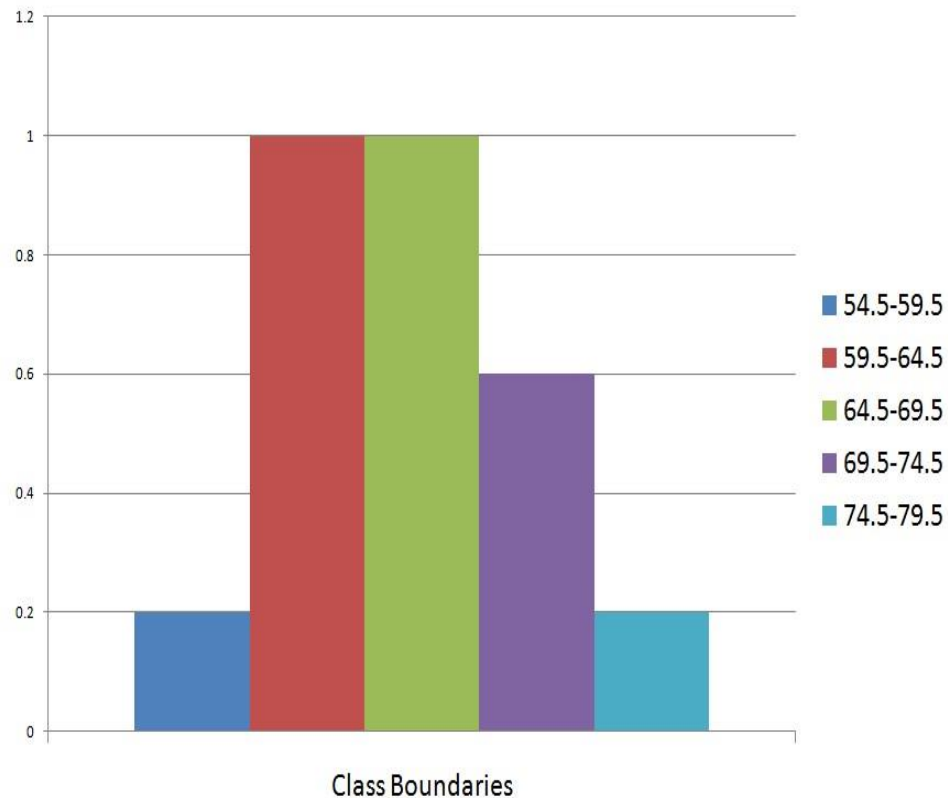
- Definition: A Bar chart which represents continuous data is called a **histogram** if
 - The bars have no spaces between them (though there may be bars with zero height, which could look like spaces).
 - The area of each bar is proportional to the frequency.
 - If all the bars have the same width, then the height is proportional to the frequency.
- Example: Height ranges of students in centimeters.



- Definition: **Frequency Density** is defined as the ratio between the frequency of a class and the class width. i.e. $\text{Frequency Density} = \text{Frequency} / \text{Class Width}$
- Example:

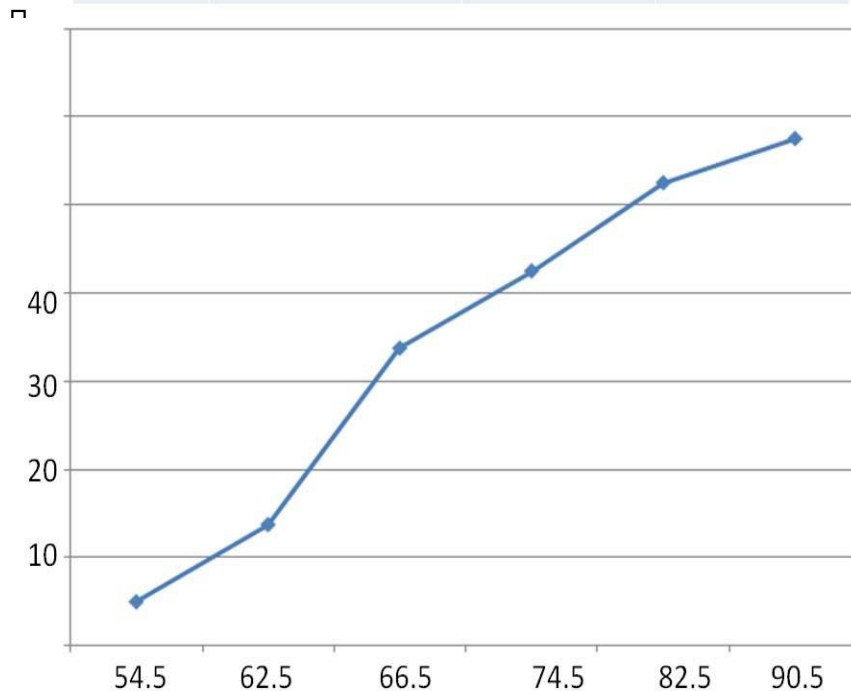
Height Range	Class Width	Frequency	Frequency Density
54.5 – 59.5	5	1	0.2
59.5 – 64.5	5	5	1.0
64.5 – 69.5	5	5	1.0
69.5 – 74.5	5	3	0.6
74.5 – 79.5	5	1	0.2

- Histogram: Frequency Density of height ranges of students



- Cumulative Frequency Graphs: Another way of representing continuous data is to draw a cumulative frequency graph. The cumulative frequencies are plotted against the upper class boundaries of the corresponding class
- Example: The grouped frequency distribution in the following table summarizes the masses in kilograms, measured to the nearest kilogram, of a sample of 38 students. Represent the data using a cumulative frequency graph

Mass (g)	Class Boundaries	Frequency	Cumulative Frequency
47 - 54	46.5 – 54.5	4	4
55 - 62	54.5 – 62.5	7	11
63 - 66	62.5 – 66.5	8	19
67 - 74	66.5 – 74.5	7	26
75 - 82	74.5 – 82.5	8	34
83 - 90	82.5 – 90.5	4	38



- We can use the cumulative frequency graph to get other information also. For example, what proportion of the students has mass less than 60 kg?

From the graph the frequency is about 8.8. The proportion under 60 kg = $8.8/38 = 0.23$ or 23%

- Pie Charts: Pie charts are useful for representing percentage allocation data.

For example, the budget of a household can be represented effectively through a pie chart. Pie charts can be thought of as **circle graphs**. To calculate the quantities represented by each slice of the pie chart we need to take the angular fraction of the given total. i.e. If a particular frequency is x , and n is the total number, then the allocation angle for that class will be $(x/n) \cdot 360$

- Example: In a survey, 90 people were asked to indicate which one of five musical instruments they played. The information is given in the following

Instrument	Number of People
Guitar	35
Violin	10
Keyboard	25
Drum	5
Recorder	15

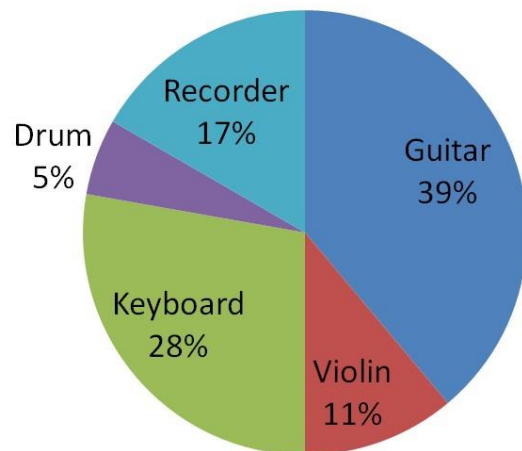
table.

Represent the data on a pie chart

Solution: We first calculate the allocation angle for each group:

Instrument	Number of People	Allocation Angle	Percentage
Guitar	35	140°	39%
Violin	10	40°	11%
Keyboard	25	100°	28%
Drum	5	20°	5%
Recorder	15	60°	17%
TOTAL	90	360°	100%

People and Musical Instruments they Play



Measures of Central Tendency

Topic 10

□ Focus of Statistics:

We are concerned in statistics with three things:

- **Measures of Location:** where the data is clustered or centered. Also called measures of central tendency.
- **Measures of Dispersion:** How the data is spread out from the center.
- **Measures of Shape:** How the data sways and peaks.

- Measures of Central Tendency: The **central tendency** is measured by averages. These describe the point about which the various observed values cluster. In mathematics, an **average**, or **central tendency** of a data set refers to a measure of the "middle" or "expected" value of the data set.
- We will be studying the following measures of central tendency: Arithmetic Mean, Geometric Mean, Weighted Mean, Harmonic Mean, Median, and Mode.
- Definition: The **Arithmetic Mean** is the sum of the observation values divided by the number of observations. It is the most widely used measure of central tendency, and is the *only measure* where the *sum of the deviations* of each value from the mean is always *zero*. The formula for

calculating the arithmetic mean of n values is: $\bar{x} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n}$ or, $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$

- Arithmetic Mean of Grouped Data: If $z_1, z_2, z_3, \dots, z_k$ are the mid-values and $f_1, f_2, f_3, \dots, f_k$ are the corresponding frequencies, where the subscript 'k' stands for the

number of classes, then the mean is $\bar{z} = \frac{\sum f_i z_i}{\sum f_i}$.

- Example: The math exam scores of 9 students are given below: 59, 66, 65, 74, 71, 67, 71, 62, 77.

59 66 65 74 71 67 71 62 77

77612

The mean score is: $\bar{x} = \frac{77612}{9} = 86.23$

- Definition: The **Geometric Mean** is often used when calculating investment returns over multiple periods, or to find a compound growth rate. It is computed by taking the n th root of the product of n values. In general, when we are dealing with numbers that need to be multiplied, the geometric mean gives a more realistic picture than the arithmetic mean. The geometric mean for n values is:

$$GM = \sqrt[n]{a_1 a_2 \dots a_n} = \left(\prod_{i=1}^n a_i \right)^{1/n}$$

- Geometric Mean of Grouped Data: If the “n” non-zero and positive variable values $x_1, x_2, x_3, \dots, x_n$ occur $f_1, f_2, \dots, f_n$ times, respectively, then the geometric mean of the set of

$$x_1, x_2, \dots, x_n \text{ where } N = \sum_{i=1}^n f_i$$

observations is defined by: $GM = \sqrt[n]{x_1 x_2 \dots x_n}$

$$= \sqrt[n]{\prod_{i=1}^n x_i^{f_i}}$$

- Example (a): Company A has grown over the last 3 years by 10 million, 12 million and 14 million dollars. What is the average annual growth amount?

This is arithmetic mean = $(10 + 12 + 14)/3 = 12$ million dollars. • Example (b):

The profit of Company B has grown over last three years by 2.5%, 3%, and 3.5%.

What is the average growth rate?

This is geometric mean = $[(1.025)(1.030)(1.035)]^{1/3} = 2.9992\%$

- Definition: The **Weighted mean** is a special case of the mean that allows different weights on different observations. Formally, the weighted mean of a non-empty set of data, $x_1, x_2, \dots, x_n$, with non-negative weights $w_1, w_2, \dots, w_n$, is the quantity calculated by

$$\bar{x} = \frac{w_1 x_1 + w_2 x_2 + \dots + w_n x_n}{w_1 + w_2 + \dots + w_n}$$

Note: The arithmetic mean is weighted mean where all the weights equal 1

- Example: Given two school classes, one with 20 students, and one with 30 students, the grades in each class on a test were:

Morning class = 62, 67, 71, 74, 76, 77, 78, 79, 79, 80, 80, 81, 81, 82, 83, 84, 86, 89, 93, 98

Afternoon class = 81, 82, 83, 84, 85, 86, 87, 87, 88, 88, 89, 89, 89, 90, 90, 90, 90, 91, 91, 91, 92, 92, 93, 93, 94, 95, 96, 97, 98, 99

What is the average score of all the students if the average for the morning class is 80 and the average of the afternoon class is 90?

Solution: The straight average of 80 and 90 is 85, the mean of the two class means. However, this does not account for the difference in number of students in each class, and the value of 85 does not reflect the average student grade (independent of class). The average student grade can be

obtained by either averaging all the numbers without regard to classes as $\bar{x} = \frac{4300}{50} = 86$, or weighting the class means by the number of students in each class: $\bar{x} = \frac{20(80) + 30(90)}{20 + 30} = 86$.

- Definition: The **Harmonic mean** is often used by investors to find the average cost of shares purchased over time. In certain situations, especially many situations involving *rates* and *ratios*, the harmonic mean provides the truest average. Formally, for a set of positive data values, $x_1, x_2, \dots, x_n$, the harmonic mean is the reciprocal of the arithmetic mean of the reciprocals:

$$HM = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}} = \frac{n}{\sum_{i=1}^n \frac{1}{x_i}}$$

- Example (c): You go on a 100 km trip. Suppose you travel half the **time** at 40 km/h and half the time at 60 km/h. What is the average speed?

This is Arithmetic mean = $(40 + 60)/2 = 50$ km/hr

- Example (d): You go on a 100 km trip. Suppose you travel half the **distance** of your trip at 40 km/h, and the remaining half at 60 km/h. What is the average speed?

This is Harmonic mean = $2/(1/40 + 1/60) = 48$ km/hr

- Definition: The **median** is the middle value of the observations such that the number of observations above it is equal to the number of observations below it. If the number of values, n is odd, then the median is the middle value, i.e. $M_e = x_{\frac{n+1}{2}}$; if n is even, then the median is the

$$\frac{1}{2}(n+1)$$

$$\frac{1}{2}(n+1)$$

average of the middle two values, i.e. $M_e = \frac{x_{\frac{n}{2}} + x_{\frac{n}{2}+1}}{2}$

$$\frac{x_{\frac{n}{2}} + x_{\frac{n}{2}+1}}{2}$$

- Median of Grouped Data: For grouped data the median is given by: $M_e = L_o + h \frac{\frac{n}{2} - F_o}{f_o}$,
where L_o = Lower class boundary of the median class $\circ$

h = Width of the median class $\circ$ f_o = Frequency of

the median class $\circ$ F = Cumulative frequency of the

pre-median class

- Steps to Find the Median of Grouped Data:

- Compute the less than type cumulative frequencies. ○

Determine $n/2$, one-half of the total number of cases.

- Locate the median class for which the cumulative frequency is more than $N/2$. ○ Determine the lower limit of the median class. This is L_0 . ○ Sum the frequencies of all classes prior to the median class. This is F . ○ Determine the frequency of the median class. This is f_0 .
- Determine the class width of the median class. This is h . ○

Apply the Formula.

- Definition: **Mode** is the value of a distribution for which the frequency is maximum. In other words, mode is the value of a variable, which occurs with the highest frequency. •

Example: The mode of the list (1, 2, 2, 3, 3, 3, 4) is 3. The mode is not necessarily well defined.

The list (1, 2, 2, 3, 3, 5) has the two modes 2 and 3.

- Mode of Grouped Data: The formula for finding mode from grouped data is

$$M_0 = L_1 + \frac{f_1 - f_0}{f_1 - f_0 + f_1 - f_2} h, \text{ where}$$

- L_1 = Lower boundary of modal class ○ f_1 = difference of frequency between modal class and class before it ○ f_2 = difference of frequency between modal class and class after
- h = class interval • Example: A commuter who travels to work by car has a choice of two different routes, V and W. He decides to compare his journey times for each route. So, he records the journey times, in minutes, for 10 consecutive working days, for each route. The results are:

Route V	53	52	48	51	49	47	42	48	57	53
Route W	43	41	39	108	52	42	38	45	39	51

- Calculate the means and medians for the two routes respectively.
- Which average do you think is more suitable for comparing the time taken on each route?
- Solution: For Route V: Mean = $(53+52+48+51+49+47+42+48+57+53)/10 = 500/10 = 50$
Arranging in ascending order, we get: 42, 47, 48, 48, 49, 51, 52, 53, 53, 57. So, the Median is $(49 + 51)/2 = 50$

For Route W: Mean = $(43+41+39+108+52+42+38+45+39+51)/10 = 498/10 = 49.8$. Arranging in ascending order, we get: 38, 39, 39, 41, 42, 43, 45, 51, 52, 108. So, the Median is $(42 + 43)/2 = 42.5$

Route W is quicker.

- Example: Consider the two sets of data A and B

A: 48, 52, 60, 60, 60, 68, 72
B: 0, 10, 60, 60, 60, 110, 120

For both sets of data the mean = median = mode = 60. Are the two data sets the same?

- If you were given nothing but the measures of location, you might be tempted to think that the two sets of data are similar. But B is much more spread out than A. It is necessary for us to devise some new measures to summarize the **spread of data**.

Measures of Dispersion

Topic 11

- The Measures of Dispersion we will consider are:
 - Range
 - Inter Quartile Range ○ Mean Absolute Deviation ○ Variance and Standard Deviation.
- Definition: The **Range** is defined as the difference between the largest and the smallest data values. i.e. Range = Largest Value – Smallest Value
 - In the previous example, the range for A is: $72 - 48 = 24$ and the range for B is: $120 - 0 = 120$.

- Limitations of Range: Range is limited as it does not give the pattern of distribution in the middle.
 - Example:

Dataset C: 2, 3, 4, 6, 8, 9 10

Dataset D: 2, 5, 6, 6, 6, 7, 10

Both the datasets have same mean, same median and same range. But the data patterns are different. In D, data is clustered around the center, but in C it is farther away from the center.

We need more accurate measures of dispersion.

- Definition: The **quantiles** are values which divide the distribution such that there is a given proportion of observations below the quantile. It is mostly used to compare one's results with others.
- Commonly used quantiles are:
 - Quartiles: The distribution is divided into quarters.
 - Quintiles: The distribution is divided into fifths.
 - Deciles: The distribution is divided into tenths.
 - Percentile: The distribution is divided into hundredths (percents).
- At a given percentile, y , with n data points **sorted in ascending order**, the position of the observation: $L_y = (n + 1) (y/100)$. Then from the dataset we can obtain P_y which is the L_y th value
 - Example, For 11 observations, the third quartile (point below which 75% of observations lie) can be identified as the 9th value since $L_y = (11 + 1) (75/100) = 9$
- Example: Consider the data set: 47 35 37 32 40 39 36 34 35 31 44
 - Find the 75th percentile point.
 - Find the 1st quartile point.
 - Find the 5th decile point.

Solution: First arrange the data in ascending order: 31, 32, 34, 35, 35, 36, 37, 39, 40, 44, 47

○ Location of the 75th percentile is the $L_{75} = (11 + 1) (75/100) = 9^{\text{th}}$ value. i.e. $P_{75}=40$ ○

Location of the 1st quartile is the $L_{25} = (11 + 1) (25/100) = 3^{\text{rd}}$ value. i.e. $P_{25}=34$ ○

Location of the 5th decile is the $L_{50} = (11 + 1) (50/100) = 6^{\text{th}}$ value. i.e $P_{50}=36$.

- Definition: Recall, the first quartile, denoted by Q_1 , is the value below which 25% of the values lie, while the third quartile, denoted by Q_3 , is the value below which 75% of the values lie. The **inter-quartile range, IQR**, is defined as $Q_3 - Q_1$.

IQR contains information about the spread of the middle 50% of the data.

Note: the second quartile, Q_2 = Median

- Another way to find Quartiles:
 - Case1: If there are even number of data values.
 - Arrange the data in ascending order
 - Split the data into upper half and lower half
 - The median of the upper half is Q_3 and the median of the lower half is Q_1
 - Case2: There are odd number of data values.
 - Arrange the data in ascending order
 - Find the median Q_2 and delete it from the list to get even number of values.
 - Use the method of case one to find Q_1 and Q_3
- How to Read the Quartiles from a cumulative graph: Results from two traffic surveys with 50 cars each taken at two observation points A and B are summarized below.

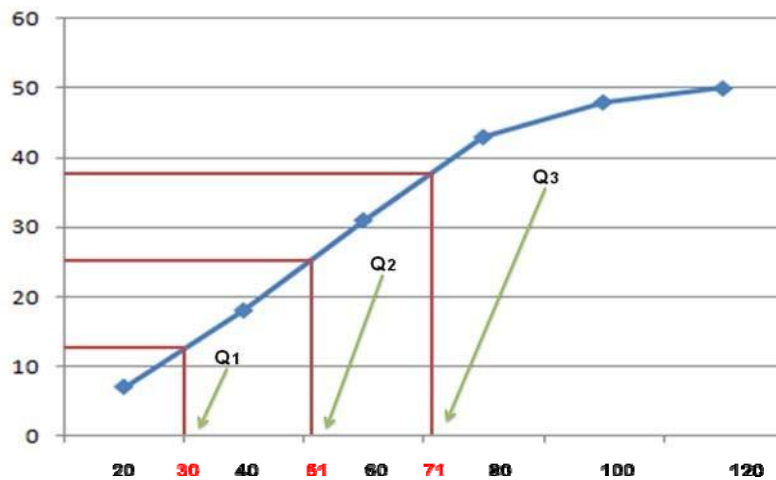
Speed	A Frequency	B Frequency	A C.Frequency	B C.Frequency
$0 \leq x < 20$	7	1	7	1
$20 \leq x < 40$	11	3	18	4
$40 \leq x < 60$	13	5	31	9
$60 \leq x < 80$	12	20	43	29
$80 \leq x < 100$	5	18	48	47
$100 \leq x < 120$	2	3	50	50
TOTALS	50	50		

Find the Median and the IQR

- Solution: Q_1 is the value below which 25% of the values lie. i.e. it is the value corresponding to $CF = 0.25(50) = 12.5$

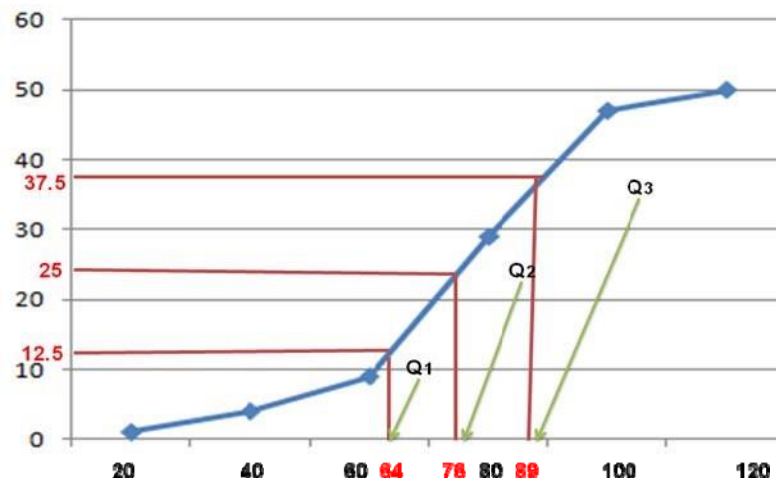
Q_3 is the value below which 75% of the values lie. i.e. it is the value corresponding to $CF = 0.75(50) = 37.5$

Q_2 is the value below which 50% of the values lie. i.e. it is the value corresponding to $CF = 0.5(50) = 25$ For A:



Median = 51, $Q_1 = 30$, $Q_3 = 71$, IQR = 41

For B:



Median = 76, $Q_1 = 64$, $Q_3 = 89$, IQR = 25

Median speed at A is lower than the median speed at B, and the IQR at A is greater than the IQR at B.

So, at A, the cars go more slowly and there is greater variation in their speeds

- Five Number Summary: One helpful way to summarize data is to give values which provide essential information about the data set. One such summary is called the **five-number**

summary. This summary gives the median, Q_2 , the lower quartile, Q_1 , the upper quartile, Q_3 , the minimum value, m , and the maximum value, M .

- **Box and Whiskers Plot:** The five-number summary can be converted into a useful diagram, called the **box-and-whisker** plot, or a **boxplot**. Steps are:
 - Draw a scale (horizontal or vertical)
 - Above the scale draw a box (rectangle) in which the left side is above the point corresponding to Q_1 and the right side is above the point corresponding to Q_3 . Then mark a third line inside the box above the point corresponding to Q_2 .
 - Draw the two whiskers: the left whisker extends from Q_1 to m , and the right whisker extends from Q_3 to M .

- **Example:** The data below give the number of fish caught each day over a period of 11 days by a fisherman. Give a five number summary of the data, and plot a box and whiskers diagram:
0, 2, 5, 2, 0, 4, 4, 8, 9, 8, 8

Solution: Rearrange the data in ascending order: 0, 0, 2, 2, 4, 4, 5, 8, 8, 8, 9

The median value is the 6th value i.e. $Q_2 = 4$

Lower quartile is the 3rd value, i.e. $Q_1 = 2$

Upper quartile is the 9th value, i.e. $Q_3 = 8$

The five-number summary is then: $m = 0$, $Q_1 = 2$, $Q_2 = 4$, $Q_3 = 8$, $M = 9$

Here is the box-and-whisker plot for the distribution of the number of fish caught over 11 days



- So far we have only looked at dispersion measuring methods that measure ranges of values. We need dispersion measures that measure variation for **each value** in the dataset.
- Definition: **Mean absolute deviation** (MAD) is the average of the absolute values of the

$$\frac{1}{n} \sum_{i=1}^n |X_i - \bar{X}|$$

deviations of individual observations from the arithmetic mean. $MAD = \frac{1}{n} \sum_{i=1}^n |X_i - \bar{X}|$

Example:

X	$X - \bar{X}$	$ X - \bar{X} $
49	-9	9
56	-2	2
55	-3	3
68	10	10
61	3	3
57	-1	1
61	3	3
52	-6	6
63	5	5
Total	0	42

$$\text{M.A.D.} = 42/9 = 4.66$$

- Limitations of MAD
 - Absolute values are not very easy to manipulate algebraically.
 - Its graph is not “smooth” and that causes problems in dealing with it in calculus.
- Definition: The **Variance** of a dataset is the mean of the squared deviations from the mean. It

$$\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$

is calculated using the formula: $\text{Var } X = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$

Standard deviation is the positive square root of the variance. i.e.

$$SD = \sqrt{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2}$$

- It can be shown through algebraic means that the variance formula can be written as

$$\text{Var } X = \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2$$

- Example:

X	$X - \bar{X}$	$(X - \bar{X})^2$	X^2
49	-9	81	2401
56	-2	4	3136
55	-3	9	3025
68	10	100	4624
61	3	9	3721
57	-1	1	3249
61	3	9	3721
52	-6	36	2704
63	5	25	3969
Total	0	274	30550

Using the definition: $\sigma^2 = \frac{1}{n} \sum (X_i - \bar{X})^2 = \frac{1}{9} (274) = 30.44$

Using the alternative formula:

$$\sigma^2 = \frac{1}{n} \sum X_i^2 - \left(\frac{\sum X_i}{n} \right)^2 = \frac{1}{9} (30550) - \left(\frac{558}{9} \right)^2 = 3394.44 - 3364 = 30.44$$

- Example: 12 boys and 13 girls, in a class of 25 students were given a test. The mean and SD of the 12 boys were 31 and 6.2 respectively; the mean and SD of the 13 girls were 36 and 4.3 respectively. Find the mean marks and SD of the whole class of 25 students.

Solution: Let the Boys' marks be represented by $x_1, x_2, \dots, x_{12}$. Let the Girls' marks be represented by $y_1, y_2, \dots, y_{13}$.

We are given $\bar{x} = \frac{1}{12} \sum x_i = 31$. This means that $\sum x_i = (31)(12) = 372$. Given that SD of Boys

is 6.2, the variance of Boys = 38.44. This means that $\frac{1}{12} \sum x_i^2 = 38.44$. This gives

$$\sum x_i^2 = 12(38.44 + 31^2) = 11993.28$$

We are also given $\bar{y} = \frac{1}{13} \sum y_i = 36$. This means that $\sum y_i = (36)(13) = 468$

Given that SD of Girls is 4.3, the variance of Girls = 18.49. This means $\frac{1}{13} \sum y_i^2 = 18.49$, 13

which gives $\sum y_i^2 = 13(18.49 - 36)^2 = 17088.37$

$$\sum x^i = y^i = \frac{372 - 468}{25} = \frac{840}{25}$$

The overall mean is $\bar{x} = \frac{\sum x^i}{n} = \frac{840}{25} = 33.6$

The overall variance is $\frac{\sum x_i^2 - \frac{(\sum x_i)^2}{n}}{n} = \frac{11993.28 - \frac{(840)^2}{25}}{25} = \frac{17088.37 - 11993.28}{25} = 34.306$

So, the overall SD = $\sqrt{34.306} = 5.86$

- Variance for Grouped Data: The formula for variance of grouped data is:

$$\frac{\sum x^2 f^2}{n} - \bar{x}^2$$

$$\text{Variance} = \frac{\sum x_i^2 f_i}{n} - (\bar{x})^2$$

- Definition: If the mean of a sample statistic is equal to the population parameter, the sample statistic is called an **unbiased estimator** of the population parameter. The sample variance becomes unbiased if we divide by $n - 1$ instead of n .

- Estimate of Population Variance: When a sample is used to actually estimate the population variance there is a different formula for the sample variance and the sample standard deviation

The unbiased estimator of the population variance is $s^2 = \frac{1}{n-1} \sum (X - \bar{X})^2$

And the unbiased estimator of the population standard deviation is $s = \sqrt{\frac{1}{n-1} \sum (X - \bar{X})^2}$

- Chebyshev's Theorem: For any set of observations, the proportion of the observations within k standard deviations of the mean is at least: $1 - (1/k^2)$ for all $k > 1$.
- Chebyshev's inequality can be used to measure maximum amount of dispersion, regardless of the shape of the distribution.
- Example: A distribution has a mean of 3. If 75% of all of its observations lie between 1.25 and 4.75, what is the standard deviation of this distribution?

Solution: We observe that the two values – 1.25 and 4.75 - are equidistant from the mean of

3. In other words,
 $3 - k\sigma = 1.25$ and
 $3 + k\sigma = 4.75$.

This gives us $2k\sigma = 3.5$ or $k\sigma = 1.75$

Now, by Chebyshev's theorem, we have $1 - (1/k^2) = 0.75$. This gives $k = 2$. Therefore $2\sigma = 1.75$, or $\sigma = 0.875$

- Definition: The **coefficient of variation** expresses how much dispersion exists relative to the mean of a distribution and allows for direct comparison of dispersion across different data sets. It is used in investment analysis to compare *relative risks*.

$$CV = \frac{s}{\bar{x}} = \text{standard deviation of } x / \text{average value of } x.$$

- Example: Investment A has a mean return of 7% and an std dev of 0.05. Investment B has a mean return of 12% and a std dev of 0.07. Which is riskier?

Solution: A's CV is $0.05 / 0.07 = .714$; B's CV is $0.07 / 0.12 = 0.583$

A has 0.714 units of risk for each unit of return while B has 0.583 units of risk for each unit of return. A has more risk per unit of return.

Skewness and Kurtosis

Topic 12

- Definition: The first four **central moments about the mean** are defined as follows: ○ μ_1

$$\mu_1 = \frac{\sum (x - \bar{x})}{n}$$

$$\mu_2 = \frac{\sum (x - \bar{x})^2}{n}$$

$$\mu_3 = \frac{\sum (x - \bar{x})^3}{n}$$

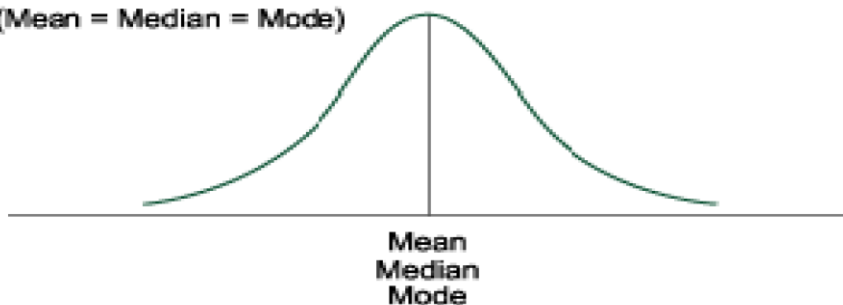
$$\mu_4 = \frac{\sum (x - \bar{x})^4}{n}$$

- Definition: **Skewness, or skew**, refers to the extent to which a distribution is **not symmetrical**. Non-symmetrical distributions may be either **positively or negatively skewed** and result from the occurrence of outliers in the data set.
 - A **positively skewed distribution** is characterized by many **outliers in the upper region, or right tail**. A positively skewed distribution is said to be skewed right because of its relatively **long upper (right) tail**.

- A **negatively skewed distribution** has a disproportionately large amount of **outliers that fall within its lower (left) tail**. A negatively skewed distribution is said to be skewed left because of its **long lower tail**.
- Definition: The **moment coefficient of skewness**, β_1 , is defined as μ_3^2 / μ_2^3 , where the sign of skewness is determined by the sign of β_1 and the degree of skewness is determined by the absolute value of β_1 .
- Skewness and Measures of Central Tendency: Skewness affects the location of the **mean**, **median**, and **mode** of a distribution as summarized in the following bulleted list.
 - For a **positively skewed** distribution, the **mode is less than the median, which is less than the mean**. The mean is affected by outliers; in a positively skewed distribution, there are large, positive outliers which will tend to "pull" the mean upward, or more positive.
 - For a **negatively skewed** distribution, the **mean is less than the median, which is less than the mode**. In this case, there are large, negative outliers which tend to "pull" the mean downward (to the left).

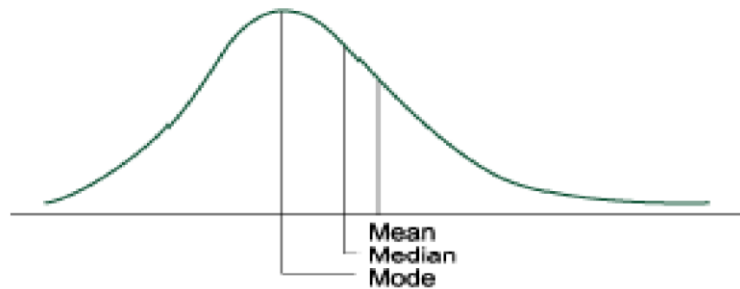
Symmetrical

(Mean = Median = Mode)

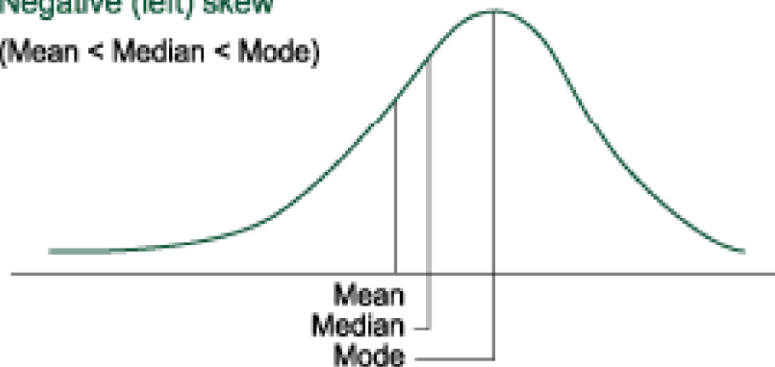


•

Positive (right) skew
(Mean > Median > Mode)



Negative (left) skew
(Mean < Median < Mode)



The diagrams illustrate an empirical relationship between mean, Median and the Mode. They suggest that the distance between the Mean and the Mode is three times the distance between the Mean and the Median, i.e. **Mean – Mode = 3(Mean – Median)**

Note: This is only an empirical and approximate relationship. There is no mathematical proof for it, and one can find counter examples where this is not true. However, in most smoothly skewed distributions this relationship holds. • Example: (a) Given Mean = 20, Mode = 15 find the value of Median. (b) Given Mode = 25, Median = 20, find the value of Mean.

Solution: (a) Mode = 3 Median — 2 Mean

$$15 = 3 \text{ Median} - (2 \times 20)$$

$$15 = 3 \text{ Median} - 40$$

$$3 \text{ Median} = 55 \text{ or Median} = 18.3$$

(b) Mode = 3 Median — 2 Mean

$$25 = (3 \times 20) - 2 \text{ Mean}$$

$$25 = 60 - 2 \text{ Mean, or } 2 \text{ Mean} = 35$$

$$\text{Mean} = 17.5$$

- Another measure of skewness is the **Pearson's coefficient of skewness**:

$$\text{Pearson's coefficient of skewness} = \frac{\text{Mean-Mode}}{\text{Standard deviation}} = \frac{3(\text{Mean-Median})}{\text{Standard deviation}}$$

- Example: Find the pearson's Coefficient of Skewness from the following data:

Income of daily Labour	Frequency (f)	(x)
40-50	3	45
50-60	5	55
60-70	10	65
70-80	8	75
80-90	4	85
90-100	4	95
100-110	1	105
Sum	35	

Solution:

Income of daily Labour	Frequency (f)	(x)	xf	x ² f
40-50	3	45	135	6075
50-60	5	55	275	15125
60-70	10	65	650	42250
70-80	8	75	600	45000
80-90	4	85	340	28900
90-100	4	95	380	36100
100-110	1	105	105	11025
Sum	35		2485	184475

Mean = 2485/35 = 71

$$\frac{1}{5}$$

$$M_0 = L_1 + h \cdot \frac{f_1}{(f_1 + f_2)} = 60 + (10) \cdot \frac{5}{5+10} = 67.14$$

$$sd = \sqrt{\frac{\sum x^2 f_i}{\sum f_i} - \left(\frac{\sum x f_i}{\sum f_i}\right)^2} = \sqrt{\frac{184475}{35} - (71)^2} = \sqrt{229.71} = 15.16$$

$$S_K = \frac{3(M_0 - \text{Mean})^2}{\text{Standard deviation}^2} = \frac{3(71 - 67.14)^2}{15.16^2} = 0.255$$

The distribution is positively skewed.

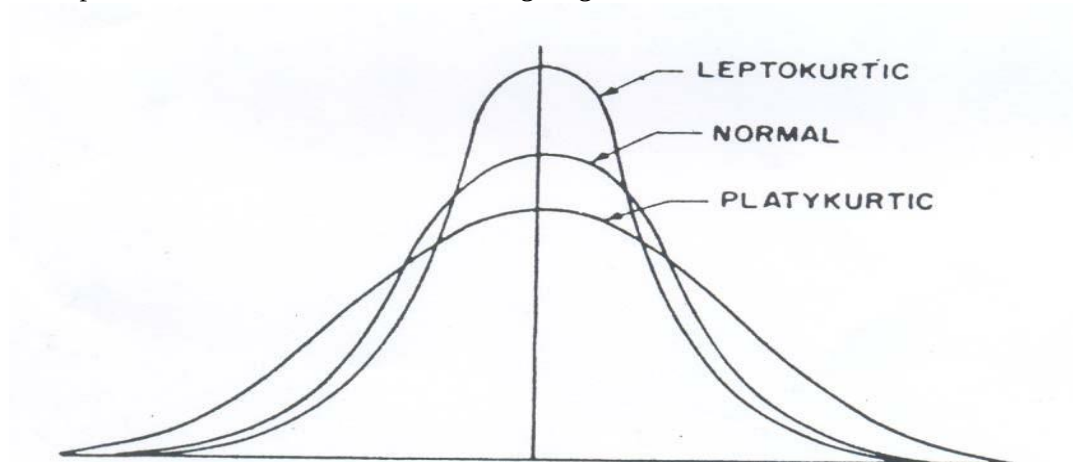
- Measuring Peakedness: **Kurtosis** is a measure of the degree to which a distribution is more or less "peaked" than a normal distribution.
- The most important measure of kurtosis based on the second and fourth moments is

$$\beta_2 = \frac{\sum (x - \bar{x})^4}{n \sum (x - \bar{x})^2} \div \frac{\sum (x - \bar{x})^2}{n} = \frac{\sum (x - \bar{x})^4}{\sum (x - \bar{x})^2} \div \frac{\sum (x - \bar{x})^2}{n}$$

distribution is **leptokurtic**. ○ If $\beta_2 \geq 3$, the

distribution is **platykurtic**. ○ If $\alpha_2 \leq 0$, the distribution is **mesokurtic**.

These pictures are illustrated in the following diagram:



Leptokurtic distribution has a **higher peak** than the normal distribution and **fatter tails** than normal distribution.

Platykurtic distribution has a **flatter peak** than the normal distribution and **thinner tails** than the normal distribution.

Probability Concepts

Topic 13

- Probability is the calculation of chance of a particular hypothesis being true or false.
- Example: Hypothesis: “Head will turn up when I toss a fair coin once”
 - The chances that this hypothesis will be true is 0.5 or 1 in 2.
 - We say that the probability that heads will show is 0.5.
- Definition: An **event** is a specified set of outcomes or a particular hypothesis.
 - We usually denote events by capital letters such as A, B, C, P, Q, R, etc.
- Notation: We denote probability by capital P, and the probability of an event A by $P(A)$.

Example: In the above example if we say that A is the event that head will turn up when we toss a fair coin once, then we write $P(A) = 0.5$
- An event can be a single outcome or a combination of outcomes. The following are all examples of events:
 - A = Head shows up when we toss a coin ○ B = Head shows up on the second and the third toss of a coin.
 - C = A number greater than 3 appears when we roll a die.
- Definition: Two events are **mutually exclusive** if they cannot happen at the same time. •

Example: When we toss a coin, the events H {= Head shows up} and T {=Tail shows up} are mutually exclusive as they both cannot happen at the same time.
- Definition: A collection of events is **exhaustive** if all the possible elementary events under the experiment are covered by the event(s) considered together. In other words, the events are said to be exhaustive when they are such that at least one of the events compulsorily occurs.

- Example: When we toss a coin, the events H and T are exhaustive as they cover all the possibilities of the experiment.
- Definition: A **Probability Space** or **Sample Space** is the set of all the possible outcomes in an experiment.
- Example: When we toss a fair coin three times, we get the following Sample Space:

□ HHH HHT HTH THH TTH THT HTT TTT, , , , , , □

- The probability of an event A is defined by: $P(A) = \frac{\text{\# of ways in which A can occur}}{\text{\# of events in the sample space}}$
- Characteristics of a Probability Distribution:
 - The probability of any event A lies between 0 and 1, i.e. $0 \leq P(A) \leq 1$
 - The sum of probabilities of all the mutually exclusive and exhaustive events in a probability distribution equals 1.
- Definition: Two events A and B are **independent** if knowledge of one event happening does not affect the probability of occurrence of the other event.
- Example: Suppose an experiment consists of tossing a fair coin twice. Let A be the event that Head turns up on the first toss and let B be the event that Tail turns up on the second toss. Then A and B are independent as the occurrence of one does not affect the occurrence of the other.
- Definition: If A is an event, then A^c (or A') is called the **complement** of A, and $P(A) + P(A^c) = 1$
- Multiplication Rule: The probability that event A **and** event B occur together is given by the multiplication rule: $P(A \cap B) = P(A) \cdot P(B) = P(A) \cdot P(B)$. ()
- Addition Rule: The probability that event A **or** event B occur is given by the addition rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- Example: The following table summarizes the availability of trucks with air bags and bucket seats at a dealership.

	Bucket Seats	No Bucket Seats	Total
Air Bags	75	50	125
No Air Bags	35	60	95
Total	110	110	220

What is the probability of selecting a truck at random that has either air bags or bucket seats?

Solution: The addition rule for probabilities is used to determine the probability of at least one event among two or more events occurring. The probability of each event is added and the joint probability (if the events are not mutually exclusive) is subtracted to arrive at the solution. $P(\text{air bags or bucket seats}) = P(\text{air bags}) + P(\text{bucket seats}) - P(\text{air bags and bucket seats}) = (125/220) + (110/220) - (75/220) = 0.57 + 0.50 - 0.34 = 0.73$ or 73 percent.

Alternative: $1 - P(\text{no airbag and no bucket seats}) = 1 - (60/220) = 73\%$

SPSS Applications Usage and all steps?

□ Originally it is an acronym of **Statistical Package for the Social Science** but now it stands for **Statistical Product and Service Solutions**

Application (Jo muji samjh aayie)

Data Editor
Output Viewer
Syntax Editor
Script Window

Usage in Research

SPSS is short for Statistical Package for the Social Sciences, and it's used by various kinds of researchers for **complex statistical data analysis**. ... Most top research agencies use SPSS to **analyze survey data and mine text data** so that they can get the most out of their research projects.

Variable View Window?

The Variable View tab **displays information about the variables in your data**. You can get to the Variable View window in **two ways**: In the **Data Editor window**, click the Variable View tab at the bottom. In the Data Editor window, in the Data View tab, **double-click a variable name at the top of the column. (Internet)**

Complementary, Dependent and independent Events?

Dependent Events: Two events A and B are dependent events when the occurrence of event A has an influence on the occurrence of an event B.

$$P(A \text{ and } B) = P(A) \times P(B/A)$$

Independent Events: Two events A and B are said to be independent events when the occurrence of an event A has no effect on the occurrence of an event B.

$$P(A \text{ and } B) = P(A) \times P(B)$$

Many times its not obvious whether two events are independent or not. In such cases we use following formula.

$$P(A/B) = P(A)$$

Complementary Events:

To every event A, there corresponds another event A^c , called the complement of A that consists of all other outcomes in the sample space not in event A. The word NOT is used to describe the complement. Since the event and its complement must account for all the **outcomes of an experiment, their probabilities must add up to one**. Complementary events are always mutually exclusive events but mutually exclusive events are not always complementary event.

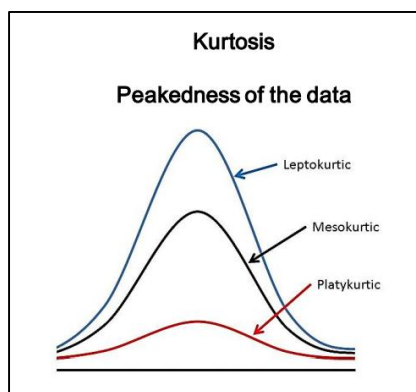
How to write name in window view?

Name

The first character of the variable name must be alphabetic □ Variable names must be unique, and have to be less than 64 characters. □ Spaces are NOT allowed. **(PPT)**

Click the Variable View tab. Type the name for your first variable under the Name column. You can also enter other information about the variable, such as the type (the default is "numeric"), width, decimals, label, etc. Type the name for each variable that you plan to include in your dataset. **(Internet)**

Draw kurtosis Graph / Draw Peakedness and Kurtosis?



Correlation?

Correlation is a statistical technique used to determine the degree to which two variables are related

Define Joint probability?

Given random variables that are defined on a probability space, the joint probability distribution for is a probability distribution that gives the probability that each of falls in any particular range or discrete set of values specified for that variable. **(Internet)**

Sample Parameter and Farm Define?

A statistic and a parameter are very similar. They are both descriptions of groups, like "50% of dog owners prefer X Brand dog food." The difference between a statistic and a parameter is that statistics describe a sample. A parameter describes an entire population.

Use of ROC CURVES?

Receiver Operating Characteristic (ROC) Curve, is a plot of the true positive rate against the false positive rate for the different possible cut points of the diagnostic test.

1. It shows the tradeoff between sensitivity and specificity (any increase in sensitivity will be accompanied by a decrease in specificity).
2. The closer the curve follows the left-hand border and then the top border of the ROC space, the more accurate the test.
3. The closer the curve comes to the 45-degree diagonal of the ROC space, the less accurate the test.
4. The slope of the tangent line at a cut point gives the likelihood ratio (LR) for that value of the test.

Percentile Calculation 75, of 41.0,8 27.1,34.7 ? ???????????

Measure of dispersion?

Dispersion (also called variability, scatter, or spread) is the extent to which a distribution is stretched or squeezed. Common examples of measures of statistical dispersion are the variance, standard deviation, and interquartile range.

Sampling Methods?

A sampling method is a procedure for selecting sample members from a population. Three common sampling methods are: simple random sampling, stratified sampling, and cluster sampling.

Step of Syntax Editor?

Click „Analyze," „Descriptive statistics," then click „Frequencies." □ Put „Gender" in the Variable(s) box. □ Then click „Charts," „Bar charts," and click „Continue." □ Click „Paste." □ Highlight the commands in the Syntax editor and then click the run icon. □ You can do the same thing by right clicking the highlighted area and then by clicking „Run Current"

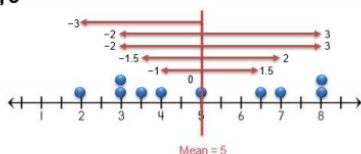
Mean Variance Formula?

Mean Absolute Deviation

Example: In a clinical study 10 breast cancer patients were followed up for the period of the time until all of them died. Their survival times (in months) are given:

2, 3, 3, 3.5, 4, 5, 6.5, 7, 8, 8

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} = \frac{50}{10} = 5$$

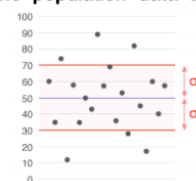


Variance

It is defined as the mean of the squared deviations of the observations from their mean.

If the variance is calculated for the population data then it is denoted by σ^2

$$\sigma^2 = \frac{\sum_{i=1}^N (X_i - \mu)^2}{N}$$



How simple stratified sample is done / chosen?

Stratified sampling refers to a type of sampling method. With stratified sampling, the researcher divides the population into separate groups, called strata. Then, a probability sample (often a simple random sample) is drawn from each group. ... Or it may be possible to increase the precision with the same sample size.

How can you Draw Height of students in descending order?

Click data, sort cases, double click 'height of students,' click 'descending,' and finally click ok.

Draw bar graph 2 to 4 population element?

?????????????

Define type of windows?

Data Editor Output Viewer

Syntax Editor

Script Window.

Output Viewer: Output Viewer Displays output and errors. Extension of the saved file will be "spv."

Script Window: Script Window Provides the opportunity to write full-blown programs, in a BASIC-like language. Text editor for syntax composition. Extension of the saved file will be “sbs.”

Syntax Editor: Syntax Editor Text editor for syntax composition. Extension of the saved file will be “sps.”

Data Editor: The Data Editor provides a convenient, spreadsheet-like method for creating and editing data files. The Data Editor window opens automatically when you start a session. The Data Editor provides two views of your data: Data View. This view displays the actual data values or defined value labels.

How third quintile is calculate?

- 1) Draw a central box (rectangle) from the first quartile to the third quartile
- 2) Draw a vertical line to mark the median
- 3) Draw horizontal lines (called whiskers) that extend from the box out to the smallest and largest observations that are not outliers
- 4) If there are any outliers, mark them separately

Regression analysis?

Regression analysis is a powerful statistical method that allows you to examine the relationship between two or more variables of interest. While there are many types of regression analysis, at their core they all examine the influence of one or more independent variables on a dependent variable.

Types of events?

Mutually Exclusive Events • Collectively Exhaustive Events • Equally Likely Events • Independent and Dependent Events.

Box and Whisker Plot Steps?

A useful visual device for communicating the information contained in a dataset. • Sometimes simply called “Box Plot”.
• Makes use of the quartiles.

Step 1: Represent the variable of interest on the horizontal axis.

Step 2: Draw a box in the space above the horizontal axis in such a way that the left end of the box aligns with the lower Quartile i.e. Q1 and the right end of the box aligns with the upper quartile i.e. Q3 .

Step 3: Divide the box into two parts by a vertical line that aligns with the median i.e. Q2 .

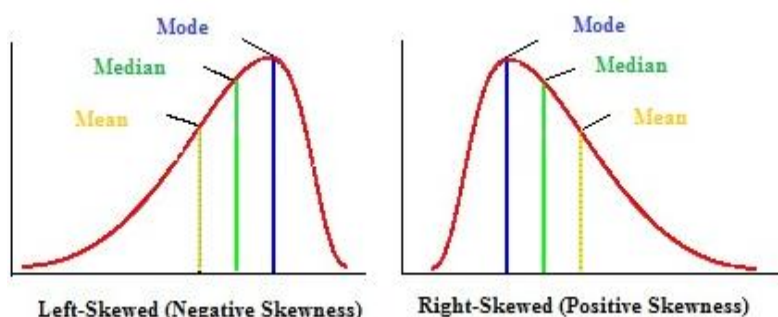
Step 4: Draw a horizontal line called whisker from the left end of the box to a point that aligns with the smallest measurement in the data set.

Step 5: Draw another horizontal line or whisker from the right end of the box to a point that aligns with the largest measurement in the data set.

Box and Whisker plot provides a very quick summary of our quantitative variable irrespective of it to be Symmetric or Skewed. Box and Whisker plot is drawn for quantitative variables only. If we want to see the distribution of a quantitative variable across different levels of a qualitative variable then Box plots provides a good and quick summary.

Show negative and Positive Skewed?

That's because there is a long tail in the negative direction on the number line. The mean is also to the left of the peak. A right-skewed distribution has a long right tail. Right-skewed distributions are also called positive-skew distributions.



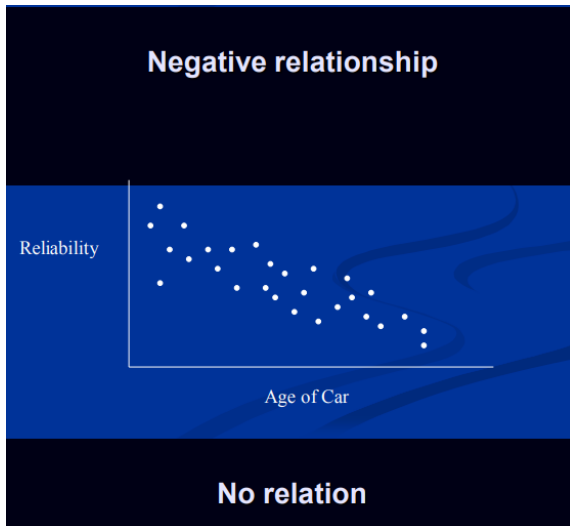
How to plot regression line?

Click „Graphs,” „Legacy Dialogs,” „Interactive,” and „Scatterplot” from the main menu. Drag „Current Salary” into the vertical axis box and „Beginning Salary” in the horizontal axis box. ☐ Click „Fit” bar. Make sure the Method is regression in the Fit box. Then click „OK”.

Variable Editor of SPSS?

??????????? In PPTS Variable View....

Show negative relationship with help of graph?



What are requirements of sample in CRT?
??????????

Differentiate metal and nonmetal according to box plot?
??????????

Explain the additive role with exclusive event?

The Additive Rule of Probability. ... We can rephrase the definition as follows: The probability that either event or event occurs is equal to the probability that event occurs plus the probability that event occurs minus the probability that both occur.

How subject help in disease patient and non-disease patient?

- o Ideally all subjects having the disease should be classified as “having the disease” and vice versa.
- o Practically, we must be aware of the fact that “Tests are not always infallible”.
- o The ability to classify individuals into the correct disease status depends on the accuracy of the tests, among other things

Explain regression equation?

Regression equation describes the regression line mathematically v Intercept v Slope (**PPTS**)

A linear regression line has an equation of the form $Y = a + bX$, where X is the explanatory variable and Y is the dependent variable. The slope of the line is b , and a is the intercept (the value of y when $x = 0$). (**Internet**)

What is PPO and YCP?

PPO = Polyphenol Oxidase activity

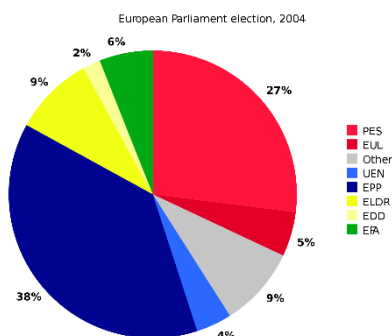
YPC = Yellow Pigment Content.

Why complimentary event are mutually exclusive but mutually exclusive are not complimentary event?

Complementary & Mutually Exclusive Events. Complementary events are two outcomes of an event that are the only two possible outcomes. This is like flipping a coin and getting heads or tails. ... Rolling a die and getting a 1 or 2 are not complementary since there are other outcomes that may happen (3, 4, 5, or 6).

Mutually exclusive events cannot happen at the same time. For example: when tossing a coin, the result can either be heads or tails but cannot be both. Events are independent if the occurrence of one event does not influence (and is not influenced by) the occurrence of the other(s).

Pie Char Draw?



$N=5$ and $P=0.60$ then find binomial trial?

PPTS (Binomial Distribution-I)

Comparison of absolute measure and relative measure variable?

There are two ways to measure numerical data. The absolute quantity is a measure of the absolute occurrence of the variable. It is a “sheer” number. ... The relative quantity is generally an absolute quantity divided by some other quantity.

99 People are salad, 30 out of which were ill by gastro disease, then probability of illness?

Measures of Morbidity - I

Example B: In an outbreak of gastroenteritis among attendees of a corporate picnic, 99 persons ate potato salad, 30 of whom developed gastroenteritis. Calculate the risk of illness among persons who ate potato salad.

Numerator = 30 persons who ate potato salad and developed gastroenteritis

Denominator = 99 persons who ate potato salad
 $10^0 = 10^2 = 100$

Risk = "Food-specific attack rate" = $(30/99) \times 100$
 $= 0.303 \times 100$
 $= 30.3\%$

Classical probability how it's difficult to draw a frame?

Classical probability is the statistical concept that measures the likelihood of something happening, but in a classic sense, it also means that every statistical experiment will contain elements that are equally likely to happen.

Responsibility of team leader during computing profession?

A team leader is responsible for guiding a group of employees as they complete a project. They are responsible for developing and implementing a timeline their team will use to reach its end goal. Some of the ways team leaders ensure they reach their goals is by

delegating tasks to their members, including themselves.

Define frame?

Frame – list of elements that sample selected from (ASW, 258). Eg. telephone book, city business directory. May be able to construct a frame.

BIO401 - Biostatistics

SHORT QUESTIONS AND MCQs FOR FINAL TERM

BY
MUHAMMAD FARYAD

SHORT QUESTIONS AND MCQs FOR FINAL TERM BIO 401

1. What are outlier?

Unusual data values as compared to the rest of the set. They may be distinguished by gaps in a histogram.

2. Name of probability sampling design?

Simple random sampling Stratified sampling □ Systematic sampling □ Cluster (area) sampling □ Multistagesampling □

3. Difference between qualitative quantative variables?

Quantitative Variables It can be measured in the usual sense. For example: - the heights of adult males,

- the weights of preschool children, - the ages of patients seen in a - dental clinic.

Qualitative Variables Many characteristics are not capable of being measured. Some of them can be ordered or ranked. For example: - classification of people into socioeconomic groups, -social classes basedon income, education, etc

4. What are shapes of distribution?

Symmetric • Data is symmetric if the left half of its histogram is roughly a mirror image of its right half.

Skewed • Data is skewed if it is not symmetric and if it extends more to one side than the other.

Uniform • Data is uniform if it is equally distributed (on a histogram, all the bars are the same

height or approximately the same height).

5. What is data and what are difference source of data?-

The raw material of Statistics is data. We may define data as figures. Figures result from the process of counting or from taking a measurement. For example: - When a hospital administrator counts the number of patients (counting). - When a nurse weighs a patient (measurement)

Sources:

Data Scales

Data are generally classified into four types:

1. **Nominal** – Categorical data
2. **Ordinal** – shows ranks, intervals may vary
3. **Interval** – intervals are constant, arbitrary 0
4. **Ratio** – Numeric data with a 'real' 0 value.

Ordinal, Interval and Ratio scales are all **Numeric** data.

What is SPSS?

Originally it is an acronym of Statistical Package for the Social Science but now it stands for Statistical Product and Service Solutions

One of the most popular statistical packages which can perform highly complex data manipulation and analysis with simple instructions

Box and Whisker Plot

- A useful visual device for communicating the information contained in a dataset.

- Sometimes simply called "Box Plot" • Makes use of the quartiles

Regression analysis

Regression analysis is a set of statistical methods used for the estimation of relationships between a dependent variable and one or more independent variables. It can be utilized to assess the strength of the relationship between variables and for modeling the future relationship between them.

Mutually Exclusive Events

Two or more events are said to be mutually exclusive if the events do not have any outcomes in common. They are events that can not occur together.

Dependent and Independent Events

Dependent Events: Two events A and B are dependent events when the occurrence of event A has an influence on the occurrence of an event B.

Independent Events: Two events A and B are said to be independent events when the occurrence of an event A has no effect on the occurrence of an event B.

Outliers

- Unusual data values as compared to the rest of the set. They may be distinguished

by gaps in a histogram.

Dot plot

- Consists of a graph in which each data value is plotted as a point along a scale of values

Time Series

- Data set is composed of quantitative entries taken at regular intervals over a period of time.

Pie Chart

- A circle is divided into sectors that represent categories

Pareto Chart

- A vertical bar graph in which the height of each bar represents frequency or relative frequency.

A boxplot is a visual representation of the five-number summary of the distribution of a numerical variable. This consists of:

- The minimum value of the distribution
- The first quartile
- The median
- The third quartile
- The maximum value of the distribution

The Mean and Median are Special cases of a family of parameters known **as location parameters**.

What is frequency distribution and frequency histogram?

Frequency Distribution • A table that organizes data values into classes or intervals along with number of values that fall in each class (frequency, f). 1. Ungrouped Frequency Distribution – for data sets with few different values. Each value is in its own class. 2. Grouped Frequency Distribution: for data sets with many different values, which are grouped together in the classes.

Frequency Histogram • A bar graph that represents the frequency distribution. • The horizontal scale is quantitative and measures the data values. • The vertical scale measures the frequencies of the classes. • Consecutive bars must touch

Other location parameters which help us to determine the position of a particular observations in a data are called **Measures of Position**

These Measures of Position when help us to divide the sample data into n equal parts, or divide the probability distribution into contiguous interval with equal probabilities are called **QUINTILES**.

Define **Dispersion**?

Dispersion is defined as the extent to which the observations in the dataset are spread away from the central value.

Define Measures of Dispersion and its type?

Measures of Dispersion are the methods that convey information regarding the amount of variability present in the set of data.

Type

i- Measures which are expressed in terms of the same units as the original data are termed as

Absolute Measures of Dispersion.

ii- Measures which are expressed in terms of the ratios and percentages these measures are independent of the units of measurement and termed as **Relative Measures of Dispersion.**

Range is defined as the difference between the maximum and the minimum value in the data.

Range = highest data value – lowest data value ; $Range = x_m - x_o$

Quartile Deviation (Q.D.)

- Quartile Deviation is the half distance between the Upper Quartile Q3 (i.e. 75th percentile) and Lower Quartile Q1 (i.e. 25th Percentile).

- Essentially describes that how much the middle 50% of the data varies.

- Also known as semi – interqu

Mean Absolute Deviation

Arithmetic Mean of the absolute deviation measured either from the mean or median.

$$\text{Mean Absolute Deviation} = \frac{\sum_{i=1}^n 1 \left[x_i - \bar{x} \right]}{n}$$

Variance

It is defined as the mean of the squared deviations of the observations from their mean.

$$S^2 = \frac{\sum_{i=1}^n 1 \left(x_i - \bar{x} \right)^2}{n}$$

Standard Deviation

The standard deviation (SD) is the square root of the variance.

$$S^2 = \sqrt{\frac{\sum_{i=1}^n 1 \left(x_i - \bar{x} \right)^2}{n}}$$

What Is Skewness?

Skewness refers to a distortion or asymmetry that deviates from the symmetrical bell curve, or normal distribution, in a set of data.

How do you interpret skewness and kurtosis?

For skewness, if the value is greater than + 1.0, the distribution is **right skewed**. If the value is less than -1.0, the distribution is left skewed. For kurtosis, if the value is greater than + 1.0, the distribution is leptokurtik. If the value is less than -1.0, the distribution is platykurtik.

WHAT IS A BOX AND WHISKER PLOT?

A box and whisker plot is defined as a graphical method of displaying variation in a set of data. In most cases, a histogram analysis provides a sufficient display, but a box and whisker plot can provide additional detail while allowing multiple sets of data to be displayed in the same graph.

WHY USE A BOX AND WHISKER PLOT?

Box and whisker plots are very effective and easy to read, as they can summarize data from multiple sources and display the results in a single graph. Box and whisker plots allow for comparison of data from different categories for easier, more effective decision-making.

WHEN TO USE A BOX AND WHISKER PLOT?

Use box and whisker plots when you have multiple data sets from independent sources that are related to each other in some way. Examples include:

- Test scores between schools or classrooms
- Data from before and after a process change
- Similar features on one part, such as camshaft lobes
- Data from duplicate machines manufacturing the same products

HOW TO MAKE A BOX AND WHISKER PLOT?

The procedure to develop a box and whisker plot comes from the five statistics below. You can also download the box and whisker plot template.

1. **Minimum value:** The smallest value in the data set
2. **Second quartile:** The value below which the lower 25% of the data are contained
3. **Median value:** The middle number in a range of numbers
4. **Third quartile:** The value above which the upper 25% of the data are contained
5. **Maximum value:** The largest value in the data set

Probability is the measure of the likelihood that an event will occur in a Random Experiment.

Probability is quantified as a number between 0 and 1, where, loosely speaking, 0 indicates impossibility and 1 indicates certainty. The higher the probability of an event, the more likely it is that the event will occur.

The Subjective definition of probability utilizes intuition, experience, and collective wisdom to assign a degree of belief that an event will occur.

Classical Probability

DEFINITION

If an event can occur in N mutually exclusive and equally likely ways, and if m of these possess a trait E , the probability of the occurrence of E is equal to m/N .

Relative Frequency

DEFINITION

If some process is repeated a large number of times, the relative frequency of occurrence of E , m/n , will be the probability of E .

Two or more events are said to be **equally Likely** if one event is as likely to occur as the other.

Two or more events are said to be **Collectively Exhaustive events** when union of the mutually exclusive events makes the entire sample space.

Dependent Events: Two events A and B are dependent events when the occurrence of event A has an influence on the occurrence of an event B.

Independent Events: Two events A and B are said to be independent events when the occurrence of an event A has no effect on the occurrence of an event B.

To every event A, there corresponds another event A^c , **called the complement** of A that consists of all other outcomes in the sample space not in event A.

Marginal Probability refers to the probability in which the numerator of the probability is a marginal total from a table and the denominator is the grand total from the table.

When probabilities are calculated with the subset of the total group as the denominator, the result is a **conditional probability**.

Two or more events are said to be **mutually exclusive** if the events do not have any outcomes in common. They are events that can not occur together.

A **diagnostic test** is used to determine the presence or absence of a disease when a subject shows signs or symptoms of the disease.

A **screening test** identifies asymptomatic individuals who may have the disease.

Receiver Operating Characteristic (ROC) Curve, is a plot of the true positive rate against the false positive rate for the different possible cut points of the diagnostic test.

Morbidity has been defined as any departure, subjective, objective, from a state of physiological, psychological well-being.

Incidence rate or person-time rate is a measure of incidence that incorporates time directly into the denominator.

A **person-time rate** is generally calculated from a long-term cohort follow-up study

A **Mortality rate** is a measure of the frequency of occurrence of death in a defined population during a specified interval.

An experiment or a trial, whose outcome can be classified as either a success or a failure, is called a **Bernoulli Trial**.

The Discrete Distribution used for such instances is called **Poisson Probability Distribution**.

A **population** is the collection of all the elements of interest.

A **sample** is a subset of the population.

Correlation is a statistical technique used to determine the degree to which two variables are related.

Multiple regression analysis is a straightforward extension of simple regression analysis which allows more than one independent variable.

1) Name two types of statistics.....

i-Discriptive ii- inferential

2) Name two types of Discriptive statistics.....

Graphical measure

Numerical measure

3) Name two types of Numerical measure.....

Frequency distributionSummary

measure

4) Name two types of summary measure.....

Measure of central tendencyMeasure

of dispersion

5) Name three types of Measure of tendency.....

Mean

Median

Mode

The extent to which the observation in the data set are spreadaway from the central value is called....

Dispersion

1) are the method that convey information regarding the amount of variability present in the set of data.

Measure of dispersion

2) Variation, Spread, and Scatter are the term used with dispersion.

Synonymusly

3) PPO is the abrivation of.....

Polyphenolic Oxidase

4) YPC is the abrivation of.....

Yellow Pigment Content

5) How many types of measure of dispersion?

Two

6) Name two types of measure of dispersion.....

Relative measure of dispersion Absolute

measure of dispersion

7) Measures which are expressed in term of same unit as original data are termed as.....

Absolute measure of dispersion

8) Measures which are expressed in term of ratios and %age are termed as.....

Relative measure of dispersion

6) Which of the following is not the type of absolute measure of dispersion.....

- a) Range
- b) Quartile deviation
- c) **Mean**
- d) Mean absolute deviation
- e) Standard deviation and variance

7) The difference b/w minimum and maximum value in the data is called.....

Range

8) is the half difference b/w upper quartile and lower quartile?

Quartile deviation (Q.D)

9) Formula for measuring the lower quartile?

$$Q1 = 1(n+1) / 4 \text{ value}$$

10) Formula for measuring the upper quartile?

$$Q3 = 3(n+1) / 4 \text{ value}$$

11) Formula for quartile deviation?

$$Q.D = \text{Inter quartile range} / 2 = (Q3 - Q1) / 2$$

12) IQR is the abbreviation of.....

Inter quartile range

13) of the absolute deviation measured either from mean or median.

Arithmetic mean

14) is defined as the mean of the squared deviation of the observation from their mean.

Variance

15) The square root of the variance is called.....

Standard deviation

16) The variance is independent of the

Origin

17) Comparing dispersion in two data sets on the basis of standard

deviation may lead to

fallacious results.

18) C.V is the abbreviation of

Coefficient of variation

19) is defined as the departure of symmetry.

Skewness

20) What is the symbol of skewness?

S.K

21) What is the formula of skewness?

$S.K = \frac{\text{Mean} - \text{mode}}{S.D}$

22) The useful visual device for communicating the information contained in a data set is called.....

Boxplot

23) Box and whisker plot drawn for..... variable

Quantitative variables

24) Some activity with observable outcomes is called.....

Experiment

25) An experiment which produced different outcomes in multiple repetitions of the experiment, performed under similar conditions is called.....

Random experiment

26) Rolling of a dice is the example of..... experiment

Random

27) How many types of events? *two*

28) Name two types of events...

Simple and composite event

29) event consists of only one outcome..

Simple

30) If an event consists of more than one outcomes it is called..... event.

composite

31) Name various other types of events.....

Mutually exclusive event Collectively

exhaustive event Equally likely event

Independent and dependent event

32) How many types of probability?

Two

33) Name two types of probability

Objective

Subjective

34) Name further two types of objective probability....

Classical

relative frequency

35) Inyear L.J Savage give the concept of probability.

1950

36) Two or more events are said to be..... event if the event do not have any outcomes in common?

Mutually exclusive event

37) Two or more events are said to be.....event if one event is as likely to occur as the other?

Equally likely event

38) When the occurrence of event A has an influence on the occurrence of an event B then this event is called

Dependent event

39) When the occurrence of event A has no effect on event B then this event is called.....

independent event

40) Inyear the axiomatic approach to probability was formalized by the Russian Mathematician A.N Kolmogorov. *1933*

41) Three axiomatic properties of probability.....

- ***Axiom of Non — Negativity***

- **Axiom of Exhaustiveness**
- **Axiom of Additive - ness**

42) If the some of the mutually exclusive event is equal to 1 then this called.....

Axiom of Exhaustiveness

52. Formula that use to find probability that this person will be 18 year old or younger..... ?

Ans: $p(E) = \text{number of early subject} / \text{total number of subject}$

53. The probability we are seeking may be written in symbolic notation as.....

Ans: $p(E \text{ intersection } A)$

54. The symbol intersection show that.....

Joint occurrence

55. The formula of joint probability

Ans: $n(E \text{ intersection } A) / n(s)$

56. Formula of conditional probability..... is

$P(A/B) = P(A \text{ intersection } B) / P(B)$

57. Which condition show a conditional probability.....

A vertical line between A and B

When the occurrence of an event A has **no effect** on the occurrence of an event B is called

Independent Event

58. When the occurrence of event A has an **influence** on the occurrence of an event B. that is called..... ?

Dependent event

59. The way to find the probability of two event occurring at the same time is called

Multiplicative rule

60. Formula of multiplicative rule..... is?

$P(A \text{ and } E) = P(A \text{ intersection } B)$

61. Multiplicative rule is also called.....?

AND rule

62. For any **two independent event** A and E, the probability for the **joint occurrence** of the both A and E is the product of their individual probabilities..... is a

.Specific Multiplicative rule

63. Each parents has chance of making an (a) gamete

$\frac{1}{2}$

64. If the event do not have an outcomes in common.... They are the event **that cannot occur together** is?

Mutually exclusive

65. The event that occur together is.....*Not-*

mutually event

66. Following **TWO** rules use to calculate the probability anevent ?

- *.Rule of complementary event*
- *Bayes rule*

67. A..... is used to determine the presence or absence of a disease.

Diagnostic test

68. A..... identifies a symptomatic individual who may have disease.

Screening test

69. Blood pressure for hypertension is a..... ?

Screening test

70. PSA test for prostate canceris

Screening test

71. Fasting blood suger for diabetes is a.....?

Screening test

72. Blood glucose level.....

70-100mg/dL

74. Cholesterol level..... is

Less than 100mg/dL

75. Systolic blood pressure is ?

120

76. Diastolic blood pressure is...?

80

77. When test indicate a positive status.....

Having a disease

78. When test show a negative status it indicate*Not having*

disease

79. Sensitivity and specificity of a diagnosis test depend onthe ?

Quality of test

80. ROC curves is reported by the ?

Goldestien and Mushline

81. The lower the **T4 values**, show the.....
Patients are to be Hypothyroid
82. ROC analysis a part of the field called ?
Signal detection theory
83. The portion that develops disease is a?
Incidence disease
84. If you report the incidence rate of say the heart disease study asper..... person years.
2.5 , 1000
85. Prevalence is measure for following disease ?
Chronic disease such as diabetes or Osteoarthritis
- 86 is measure of frequency of occurrence of death.
Mortality
87. IS the infant mortality rate is a ratio? *Yes*
88. An experiment or a trial , whose outcome can be classified as either a success or a failure is called ?
Bernoulli trial
89. Jacob Bernoulli is a..... ?
Swiss Mathematician
90. Jacob Bernoulli was bornand death in.....year?
1705-1965
- $X=1$
When the outcome is success
92. When the outcome is failure..... ?
X=0
93. A sequence of Bernoulli trials forms a ----- ?
Bernoulli process
94. Each Bernoulli trials results in one of the ----- possible outcomes?
Two
95. The name of two possible outcomes are?
 - *Failure*
 - *Success*
96. The probability of success denoted by ----- ?
P
97. The probability of success remains ----- from trial to trial?

Constant

98. The trials are----- ?

Independent

99. A ----- results from Bernouli process?*Bionomial probability*

Distribution

100. A bionomial probability of distribution has ----- number of trials?

Fixed

101. ----- will denote the probabilities of Sucess andFailures respectively?

P and q

102. P is ----- ?

*Probability of success*103. q is

-----?

Probability of failure

104. n is a fixed number of----- ?

Trials

105. x is a----- ?

Specific number of success in n trials

106. P(x) is equal to----- ?

Probability of getting exactly x success among the n trials

107. $p^x(1-p)^{n-x}$ = is a----- ?

Probability of x success among n

108. How many methods for finding Probability?*Three*

109. In a -----, we finding the probabilities correspondingto the random variable x ?

Bionomial Distribution

110. Methods of finding probability?

- *Using the bionomial probability formula*
- *Using the Table of probabilities*
- *Using technology*

111. Suppose we have n = 5 and p= 0.60 The probabilitythat bionomial trials yeild 4 success will be given as :

Ppppq=(0.60)4 x 0.40 = 0.05184

112. **One** has to be very certain that x and p both refers tothe same cetagory being called a ----- ?

Success

113. When sampling is conducted without replacement,then consider events to be - ----- if $n < 0.05 n$:

Independent

114. The ----- suggests that values are unusual if they areoutside of these limits.

Range rule of Thumb

115. $\mu + 2\sigma$ is a -----value.

Maximum value

116. $\mu - 2\sigma$ is a ----- value.

Minimum Value

117. For $p = 0.5$ and large or small n the binomial distribution will be:

Symmetric

118. For large p and small n the binomial distribution will be -----?

Left skewed

119. For small p and small n the binomial distribution will be ___?

Right skewed

120. For small p but large n the binomial distribution approaches ?

Symmetry

121. By using MS excel we will use a function

BINOMDIST

122. In interest, number of events occurring ?

2

123. Number of Deaths claims received per day by a ----- company?

Insurance

124. Number of Alpha particles emitted from a ----- during a given period of time?

Radioactive source

125. The discrete distribution used for such instances; number of families with child, About deaths claims and alpha particles emitted from radioactive source is called ?

Poisson probability distribution

126. Who invented the poisson distribution?

French mathematician Simeon Denis Poisson

127. The probability that event occurs in a given unit of time is the ----- for all units?

Same

128. The events that occur in one unit of time is ----- of the number of events in other units.

Independent

129. The mean or expected rate is -----?

λ . (lamda)

130. λ Is----- ?

Average number of events per interval

131. The base of natural logarithms is -----?

Eulers number

132.
$$p(x) = \frac{\lambda^x}{x!} e^{-\lambda}, \quad x = 0, 1, 2, \dots$$
 the following equation is called ?

Poisson Random variable

133. The poisson process generates a----- ?

Poisson distribution

134. How many traits in poisson process ?

Three

135. In poisson process ----- are discrete??

Outcomes

136. The probability of two or more successes over a sufficiently small interval is essentially ----- ?

Zero

137. In the study drug induced of anaphyl axis among patients taking ----- as a part of their anesthesia?

Recuronium bromide

138.
$$P(X = 3) = \frac{e^{-12} 12^3}{3!} = .00177$$
 it is the----- of patients receiving recuronium?

Probability

139. poisson distribution is often used as an ----- for binomial probabilities ?

Approximation

140. Poisson distribution is important because it is often used for describing the behaviour of ----- ?

Rare events

141. A ----- is a collection of all elements of interest?

Population

142. A ----- is a subset of population?

Sample

143. kinds of sample ??

Representative or non-representative

144. ----- samples are the samples selected using the principles of probability?

Probabilistic

145. The researcher can determine the probability of any given sampling error and make ----- about population characteristics.

Statistical interferences

146. ----- sample may be reasonably representative?

Quota sample

147. Sample for each stratum or subgroup of a population is called.....

Stratified samples

148. Combinations of random, systematic, stratified and cluster sampling is called.....

Multistage sampling

149. If probability involves at each stage then which sample distribution can be obtained?

Sample statistics

150. The ----- CMAs of Regina and Saskatoon are strata?

Two

151. Lists of elements that ----- sample selected from is called ?

Frame

152. Characteristics of a population is called ?

Parameter

153. Numerical characteristics of a population is called?

Statistics

154. Probability of distribution of a static is called-----?

Sample distribution

155. In ----- each sample possible of size n has the same probability of being selected.

Simple random sample

156. It may be difficult to obtain a----- ?

Frame

157. The ----- is the frequency of occurrence of a characteristic divided by total number of elements.

Proportion

158. ----- may also be termed as population values ?

Parameters

159. A specific value of point estimator is referred as

----- of a parameter.

Point estimator

160. Difference between the point estimate and the value of parameter is called ?

Sampling Error

161. Culter poll has a sampling error of only ----- % point for the liberals.

0.2%

162. A ----- is the probability distribution for all possible values of sample static.

Sample distribution

163. In sampling distribution each sample contain ----- elements.

Different

164) ----- is a statistical technique used to determine the degree to which two variables are related.

Correlation

165) Name of ----- two quantitative variables

One variable is called independent (X) and

The second is called dependent (Y)

166) measures the nature and strength between two variables of the ----- type

Quantitative.

167) ----- The sign of ----- denotes the nature of association
 r

168) The value of ----- denotes the strength of association. r

169) ----- Statistic showing the degree of relation between two variables is also called

Pearson's correlation or product moment correlation coefficient.

170) ----- If the sign is +ve ----- this means the relation is

direct.

171) While if the sign is ----- this means an inverse or indirect relationship.
-ve

172) ----- Regression tells us how to draw the straight line described by the

correlation.

173) ----- describes the strength of a linear relationship between two variables.

Correlation

174) If $r = \dots\dots\dots$ this means no association or correlation between the two variables.

Zero

175) If $0 < r < 0.25 = \dots\dots\dots$ correlation.

weak

176) If $0.25 \leq r < 0.75 = \dots\dots\dots$ correlation.

intermediate

177) If $0.75 \leq r < 1 = \dots\dots\dots$ correlation.

strong

178) If $r = 1 = \dots\dots\dots$ correlation.

perfect

179) Spearman Rank Correlation Coefficient (r_s)

It is a non-parametric measure of correlation.

180) The names of four windows.

Data editor Output

viewer Syntax editor Script

window

181) SPSS stands for.....

Statistical Product and Service Solutions.

182) is Spreadsheet-like system for defining, entering, editing, and displaying data. Extension of the saved file will be sav.

Data Editor

183) Displays output and errors. Extension of the saved file will be spv.

Output Viewer

184) The..... shows that more people who receive wireless service tend to own PDA compared to people who do not receive wireless service.

graph

185) Text editor for syntax composition. Extension of the saved file will be sps.

Syntax Editor

186) Provides the opportunity to write full-blown programs, in a BASIC-like language. Text editor for syntax composition. Extension of the saved file will be sbp.

Script Window

187) There are two sheets in the window:

1. Data view

2. Variable view

188) The basic analysis of SPSS that will be introduced in this class.

— *Frequencies*

- Descriptives
- Linear regression analysis

- 1) Statistics is a field of study that is concerned with: collection, organization, summarization, **all of these**
- 2) The use of tools for the data that is derived from biological sciences and medicine Biological, chemical, **statistical**
- 3) Bio-statistical methods were used in the ancient civilizations like Ancient Greece, Ancient Rome, India and China **all of these**
- 4) One of the earliest known demographic / Bio-statistics study was conducted by John Graunt (1662) **(1662)**. 1663 1664
- 5) John Graunt developed life tables to warn off the onset and spread of Bubonic plague in **London**, America, Japan
- 6) Famously known as “Father of Modern Statistical sciences”. **Sir Ronald A. Fisher**. Ronald A. Fisher
- 7) Applications of Biostatistics are Genomics, proteomics, genetics **all of these**
- 8) “To understand God’s thoughts we must study statistics, for these are the measures of purpose” **HIS**, HIV HCV HBV
- 9) EXAMPLE OF CONTINUOUS VARIABLE IS HEIGHT, WEIGHT, CIRCUMFERENCE **all of these**
- 10) A Statistic: $\bar{x}$ is a descriptive measure computed from the data of a POPULATION, GROWTH, PROGRESS **NON OF THESE**
(sample)
- 11) A Parameter: It is a descriptive measure computed from the data of **a population**, SAMPLE, GROWTH
- 12) A measure of central tendency is a measure which indicates where the data is. **Middle**, UPPER LOWER
- 13) The three most commonly used measures of central tendency are: The Mean, the Median, and **the Mode**.
- 14) Here is a random sample of size 10 of ages, where $\bar{x} = (42 + 28 + \dots + 37) / 10 = 36.2, 36.3, 36.4, 36.5, \underline{36.6}$
- 15) $(n+1)/2$ th IS THE FORMULA OF MEAN MOD OR **MEDIAN**
- 16) If all values are different there is no MEAN, MEDIAN **MODE**
- 17) Sometimes, there are more than one
1. MEAN, MEDIAN **MODE**
- 18) A descriptive value for a population is called SAMPLE POPULATION **PARAMETER**
- 19) Quantitative Variable = , , S S S MEAN N M n
- 20) The Mean and Median are Special cases of a family of parameters known as?
location parameters, location sample, location technique
- 21) Other location parameters which help us to determine the position of a particular observations in a data are called **Measures of position**, sample, population
- 22) These Measures of Position when help us to divide the sample data into n equal parts, or divide the probability distribution into contiguous interval with equal probabilities are called

QUINTILES.quartiles,quality,quantity

23)cut points are also called.....?

Mode median mean **quantiles**

24)quantiles denoted by

Q Q Q k F S

25) is defined as the extent to which the observations in the dataset are spread away from the central value. **DISPERSION**,EXPANSION.DEPRESSION

26) Measures which are expressed in terms of the same units as the original data are termed as

ABSOLUTE MEASURE OF DISPERSION.COMPLETE MEASURE OF DISPERSION,
UNCOMPLETE MEASURES OF DISPERSION

27) the departure from Symmetry"IS

SKYNESS ,**SKEWNESS**

28)SKEWNESS IS DENOTED BY

SK.SL.SM

29)SKEWNESS IS GIVEN BY

Karl Pearson (1857 – 1936)

30) three axiomatic properties of Probability

Axiom of Non – Negativity Axiom of Exhaustiveness Axiom of Additive - ness

31) Box and Whisker plot is drawn for?

QUALITATIVE VARIABLE,**QUANTITATIVE VARIABLE** NON OF THESE

